

AI-Assisted Development and Deployment Report

Introduction to AI Programming
2026 Spring Semester
Assignment 4

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1. AI Tools Used

Tool: Gemini CLI

Purpose: Product requirements definition, frontend architecture design, core recommendation logic implementation, and debugging.

2. AI-Assisted Tasks

PRD Generation: Automated the creation of a structured Product Requirements Document based on a high-level service concept.

Web Implementation: Translated the markdown-based PRD into a functional React/TypeScript web application.

Algorithm Design: Developed a logical mapping system between psychological test results and curated book metadata.

UI/UX Refinement: Iteratively improved the layout and solved visual bugs in the result page.

3. Representative Prompts

Prompt 1 (Planning): "Write a clear and actionable Product Requirements Document (PRD) as a MD file. The main feature of the website is book recommendation. First, user do psychological test then, webpage recommend one book based on the result of psychological test. The result page should give enough visual contents with the reason for the recommendation. The recommendation algorithm should be based on the official book information and it should be logical."

Prompt 2 (Implementation): "md 파일을 기반으로 웹페이지를 만들어줘."

Prompt 3 (Logic Verification): "책 추천 알고리즘을 어떻게 만들었는지 설명해줘."

4. Modifications and Improvements

Visual Consistency: The AI initially attempted to fetch external book images, which resulted in mismatched or broken images on the result page.

Custom Component Implementation: To resolve the image mismatch, I instructed the AI to remove the unstable external image links and instead implement a custom CSS-based "Book Cover Component." This component dynamically renders the book title and author in a layout that mimics a real physical book cover, ensuring 100% visual accuracy and reliability.

Logic Refinement: Refined the recommendation engine to ensure the psychological axes (e.g., stress, escapism) mapped more naturally to the book themes defined in the data.json.

5. Bugs and Fixes

Issue: Book covers did not match the recommended titles.

Cause: Reliance on unpredictable external image URLs or generic placeholders.

Fix: Eliminated the image source dependency and replaced it with a stylized UI element that treats the book's metadata (Title, Author) as the primary visual identity of the cover.

6. Final Deployment

Vercel Deployment URL: <https://book-recommendation-henna.vercel.app>

Github Repository URL: https://github.com/eunseo412/book_recommendation